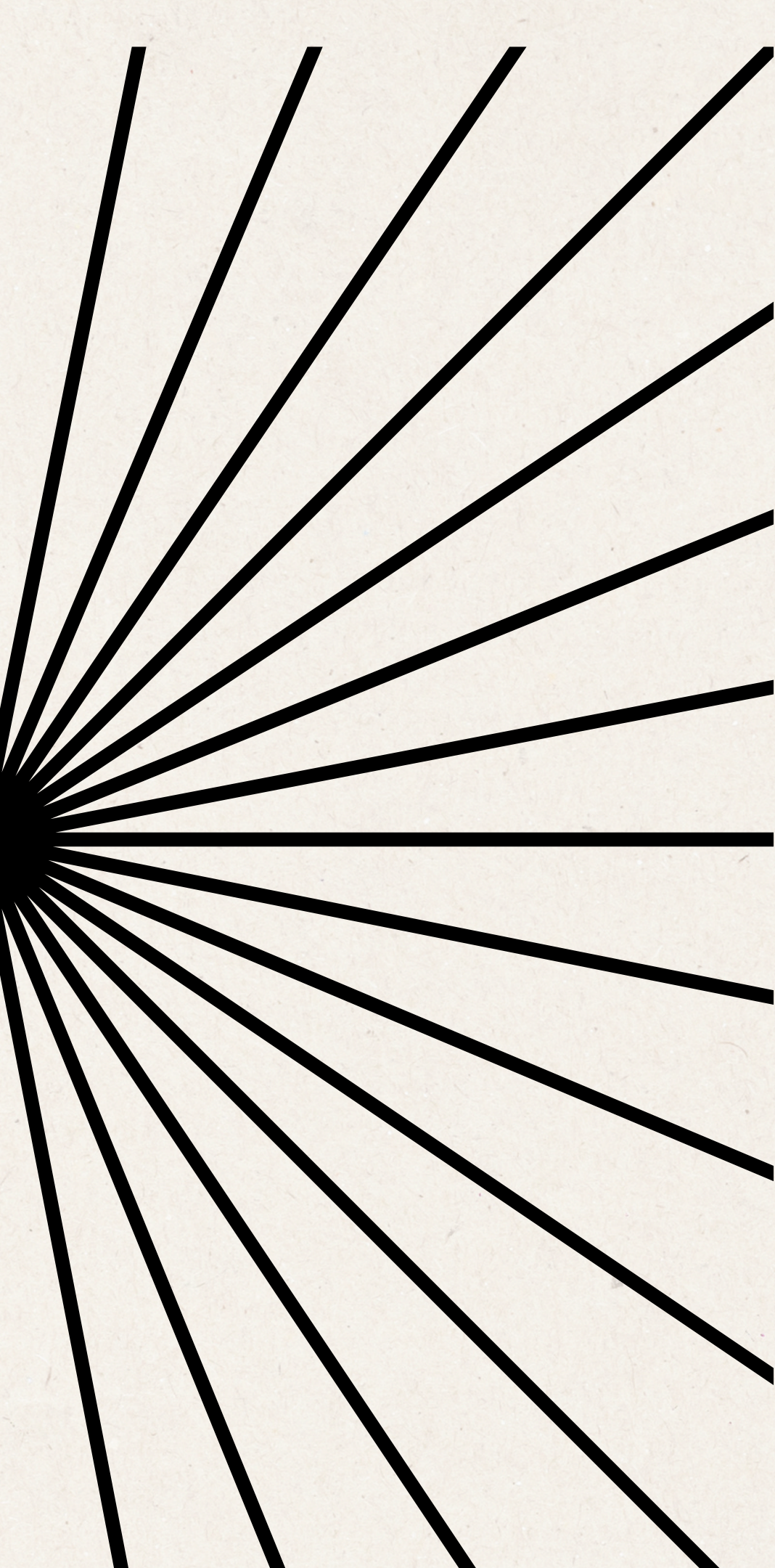




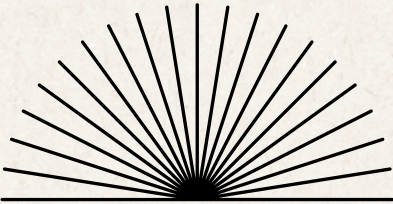
Cyfuture

FACE DETECTION

NAME OF PROJECT:
Face Detection

PRESENTED BY:
Naheed Anjum

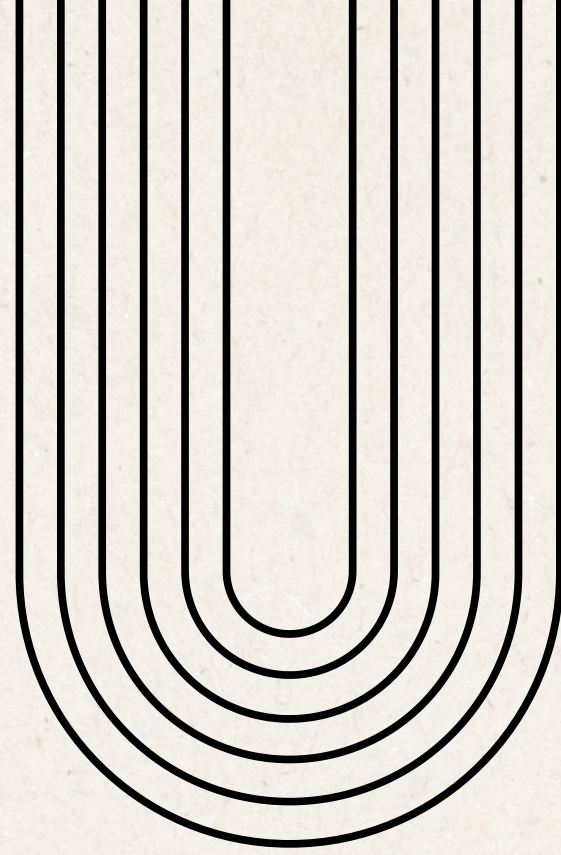
PRESENTED TO:
Cyfuture



Objective

Build a face detection system using deep learning and OpenCV.





Technology Stack

Tools & Model Details



Programming Language

Python



Libraries Used

- OpenCV (cv2) – for image processing and webcam handling
- urllib, zipfile – to fetch and extract model assets

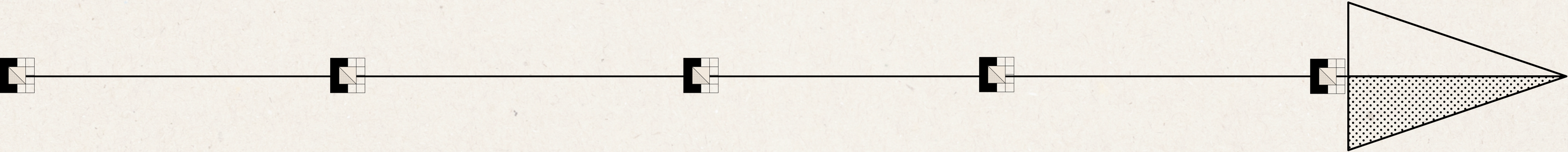
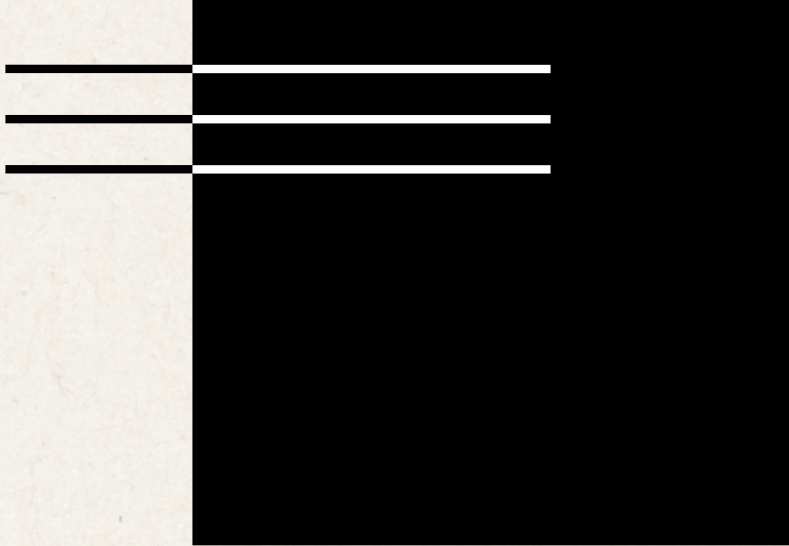


Model

- Type: Single Shot Detector (SSD) with ResNet-10 backbone
- Framework: Caffe
- Files Used:
- deploy.prototxt – defines the model architecture
- res10_300x300_ssd_iter_140000_fp16.caffemodel – pre-trained weights

Face Detection Pipeline

Briefly discuss the key dates for the project.



Capture Video

Start webcam or video capture.

Frame Preprocessing

- Resize frame to 300x300
- Convert to input blob using `cv2.dnn.blobFromImage`

Detection Output

- Load pre-trained Caffe model with OpenCV's DNN module
- Pass the blob to model

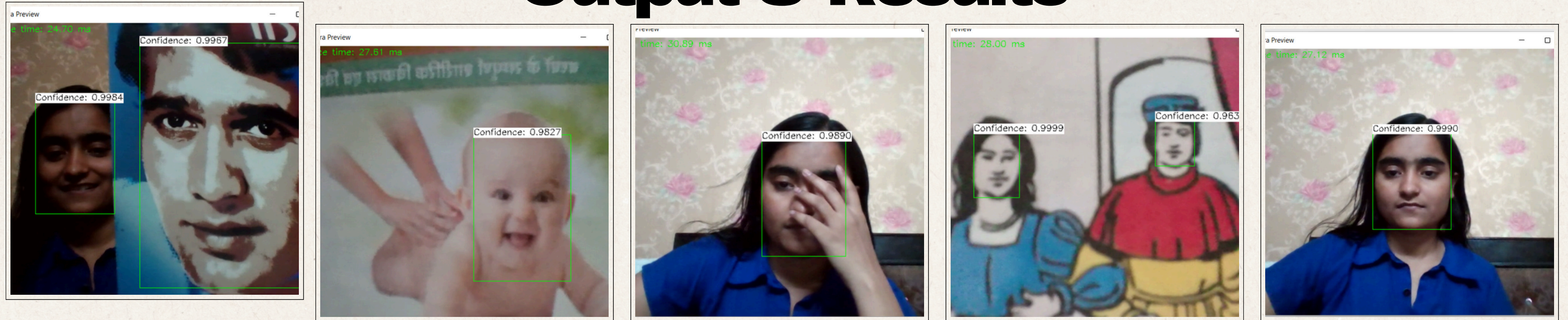
Detection Output

- Get list of bounding boxes with confidence scores
- Filter low-confidence detections

Display Output

- Draw green rectangles around faces
- Overlay confidence and inference time

Output & Results



- **Real-Time Output:** Processes each video frame and detects faces instantly
- **Confidence Score:** Each detection shows accuracy (usually >70%)
- **Inference Time:** Displays how fast each frame is processed
- **Performance:**
 - Smooth on a decent CPU (~20–30 frames per second)
 - No GPU required

Thank you
